

DoseRX

Medication & Controlled Drug Control Platform

Project overview for stakeholders, security review, and go-live planning

Prepared 06 Aug 2026 | Prototype / demonstration build

Purpose

DoseRX is a field-ready EMS medication and controlled-drug (CD) tracking system aligned to the PHECC Medication Formulary. It helps ambulance and event teams control bag stock, witness administrations and waste, keep an auditable activity trail, and give management live oversight - from QR scan on scene through to analytical reporting.

USERS

**Clinical staff +
management**

ACCESS

**PIN roles +
mobile web app**

DATA NOW

**Local demo;
cloud sync gated**

BACKEND

**Supabase
(Postgres +
Storage)**

What DoseRX does

DoseRX closes the gap between paper CD registers / bag checklists and a live digital chain of custody for medications carried on emergency and event duty.

- Sign drug bags onto and off shift with dual-PIN witness, tag/seal checks, photo evidence and optional GPS.
- Scan bag or medication QR labels to open the correct stock line, then administer or waste with dose, route and indication.
- Automatically deduct administered or wasted units from that bag's on-hand quantity (management sets stock; staff cannot overwrite counts).
- Enforce clinical-grade formulary rules (EMT / Paramedic / Advanced Paramedic) and flag out-of-scope use.
- Maintain controlled-drug visibility, discrepancy reporting, bag audits, expiry warnings and full activity history.
- Give managers dashboards, live map of bags on shift, analytics, CD register PDF and multi-page analytical reports.

1 | Capabilities & modules

Clinical / field modules

- QR | Bags & Meds - camera or manual scan; bag shift menu or medication detail to Administer.
- Administer / Waste - CAD/incident reference (no patient name), CPG-aligned dose/route/indication, witness PIN, part-dose CD support.
- Discrepancy reporting - flag count or seal issues for management follow-up.
- Offline field mode - actions queue locally when connectivity drops, then sync when back online (when cloud sync is enabled).
- Training / sandbox mode - practice on fake stock without touching live controlled-drug data.

Management / governance modules

- Dashboard & system oversight - on-shift bags, exceptions, expiries, waste and discrepancy alerts.
- Drug Bags & Stock Control - authoritative on-hand quantities; restock is management-only.
- Audit Bags - dual-witness bag audits and seal updates.
- Controlled Drugs register + CD PDF export for governance packs.
- QR Generator - unique bag QR and per-medication QR (batch, expiry, bag-specific stock line).
- Event Packs - time-boxed festival / mass-gathering bags with grade and optional CD privilege.
- Live Map - bags on shift with last-known location.
- Analytics - compliance score, activity mix, grade breakdown, medication charts, full analytical PDF.
- PHECC Formulary browser by clinical grade.

Demo bag estate (seed data)

Code pattern	Type	Role
DRX-EMT-01 / 02	Standard	EMT bags
DRX-P-01 / 02 / 03	Standard	Paramedic bags
DRX-AP-01 / 02 / 03	Standard	Advanced Paramedic bags
DRX-CD-P-01	Controlled	Paramedic CDs (e.g. Midazolam)
DRX-CD-AP-01	Controlled	AP CDs (opioids, benzos, ketamine, etc.)

Mobile-first web application

DoseRX is built as a responsive Progressive Web App (PWA-style mobile web product). Crews use it on phones and tablets in the field; managers use the same product on desktop. A separate native App Store / Play Store binary is not required for the current prototype, but the UX is designed as a mobile app workflow.

What the mobile experience provides

- Staff-first home: QR scan is the default clinical entry point on phone.
- Camera QR scanning for bag and medication labels (with manual paste / demo quick-pick for testing).
- Touch-friendly witness PIN, photo capture of seals/tags, and compact administer / waste forms.
- Responsive navigation: condensed mobile menu, alerts for exceptions, offline banner when connectivity drops.
- Works in mobile Safari / Chrome when served over HTTPS (required for camera and secure context).
- Same security model as desktop: role-gated screens (staff vs management) and witness dual-control on sensitive actions.

Field journey (mobile)

1. Sign in with staff PIN.
2. Scan bag QR -> sign out onto shift (witness + photo + tag status).
3. Scan medication QR -> review batch / expiry / qty -> Administer.
4. Enter CAD/incident number, dose, route, indication; witness verifies PIN.
5. Stock for that bag line reduces automatically; activity is logged.
6. End of duty: scan bag -> return with seal/tag photo and witness.

Future native apps (optional)

If a service requires App Store distribution, push notifications, or deeper offline hardware integration, DoseRX can later be wrapped or rebuilt as native iOS/Android clients against the same Supabase backend. The current mobile web app already proves the clinical workflow for demos and pilots.

Application controls (built today)

- Role separation: clinical staff vs management; admin-only modules (stock, audits, analytics, labels, map, events).
- PIN authentication for users; dual-PIN witness on shift sign-out/return, administration, waste and audits.
- Staff cannot overwrite stock levels - only administer/waste deductions and verification checks.
- CAD / incident labelling without patient names to reduce identifiable clinical data in the demo model.
- Photo evidence on bag movements; optional GPS on shift actions for chain-of-custody context.
- Sandbox mode isolates training data from live stock.
- Cloud write kill-switch: `VITE_SUPABASE_SYNC_ENABLED=false` keeps Try now / demos local-only (no DB writes or photo uploads) until go-live.
- Anon key only in the browser client; `service_role` key must never ship in frontend env.

Data & platform posture

- Primary store today for demos: encrypted-at-rest browser localStorage per device.
- Target production store: Supabase Postgres + private Storage bucket (evidence photos).
- Schema includes staff, bags, stock lines, shifts, activities, discrepancies and photos.
- Row Level Security (RLS) is enabled on tables; current demo policies are intentionally open for prototype wiring and must be tightened before production.
- Evidence bucket is private (not public CDN); access should be policy-bound to authenticated organisational users in production.

Production hardening checklist (before live patient/ops use)

- Replace demo PINs with proper auth (Supabase Auth / SSO / MFA) and hashed credentials - never store plaintext PINs in DB.
- Tighten RLS to organisation / role claims; remove open demo_all policies.
- Enable sync only in production env; keep marketing Try now on local or ephemeral demo tenants.
- Add audit retention, backup/restore drills, and least-privilege service keys for any server jobs.
- Complete DPIA / data-protection review for GPS, photos and operational logs under Irish/EU GDPR expectations.
- Penetration test and access reviews before any live controlled-drug operational dependency.

Important positioning

DoseRX should be presented as a working prototype / pilot platform. It demonstrates workflow and governance UX; it is not yet a certified medical device or a finished regulated CD system of record. Security and compliance claims must match the hardening state above.

4 | Supabase Pro plan - what you gain

Why upgrade from Free to Pro

The Free tier is fine for building and private demos. Once DoseRX is shared more widely, stores real evidence photos, or supports multiple crews, Supabase Pro is the practical go-live baseline for reliability, backups and capacity.

What Pro unlocks for DoseRX

Area	Why it matters for DoseRX
Daily backups + Point-in-time recovery	Recover bag stock, CD activity and audit history after mistakes or incidents
Higher database & Storage limits	Evidence photos (seal/tag images) and growing activity logs
Better compute / no aggressive pausing	Managers and field users get consistent response times
Email support & production posture	Operational issues during pilots have a vendor path
Network restrictions / security extras*	Reduce exposure of project endpoints (*check current Pro feature list)
Team access to the Supabase project	Separate builder vs ops access without sharing owner login

Recommended go-live sequence after Pro is enabled

- 1. Confirm EU region project (e.g. Ireland/Frankfurt) for data residency preference.
- 2. Apply schema v2 migration; seed real organisational staff (no demo PINs in production).
- 3. Replace open RLS with org-scoped policies; lock Storage evidence bucket.
- 4. Turn on VITE_SUPABASE_SYNC_ENABLED=true only on the production app environment.
- 5. Keep marketing site Try now either local-only or on a separate demo project with synthetic data.
- 6. Configure backups verification, monitoring alerts, and a documented restore test.
- 7. Run a controlled pilot (one station / event) before wider rollout.

Architecture note

One DoseRX web app + one Supabase project (multi-tenant by organisation when needed) is the intended model - not separate backends per customer. Mobile and desktop share that same API. Sync is currently disabled by default so public demos do not write images or clinical-operational data into the cloud until you choose to go live.

5 | Summary for decision makers

DoseRX provides

- A complete medication & CD control workflow for EMS / event medicine: scan, witness, deduct, audit, report.
- Clear separation of field users and management oversight.
- Mobile phone usability for on-scene work, plus desktop governance tools.
- Exportable evidence packs (CD register PDF, analytical PDF, QR label sets).
- A cloud path on Supabase that can be switched on when Pro and hardening are ready.

What enabling Supabase Pro enables next

- Confidence to store shared organisational data and evidence photos centrally.
- Backup / recovery suitable for operational dependence.
- Headroom for multi-crew pilots without Free-tier pause/limit surprises.
- A cleaner story for stakeholders: prototype UX now, production backend when Pro + RLS + Auth are complete.

Suggested one-line description

"DoseRX is a mobile-first medication and controlled-drug control app for EMS teams - QR bag/med workflows, dual-witness accountability, and management analytics - with Supabase ready as the secure cloud backbone once Pro and production hardening are enabled."

Document control

Product: DoseRX (LOGRX Replica codebase)

Stack: React + Vite + TypeScript + Tailwind | Supabase Postgres/Storage

Status: Active prototype - cloud sync off by default for Try now

Repo: github.com/markk0042/DoseRX

End of briefing